

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE

Course Code & Title: EC6007 Speech Processing

Name of the Faculty member: Mr.A.Azhagu Jaisudhan Pazhani

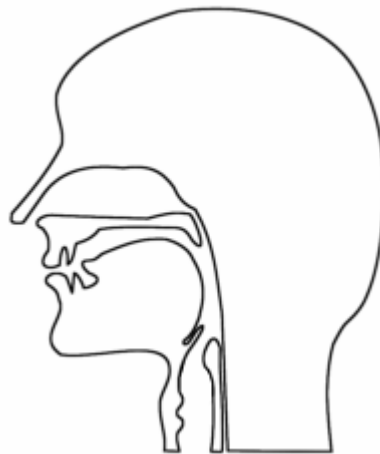
Name of the Topic: Articulatory Phonetics

Name of the Innovative Practice: Real Time Examples

UNIT I BASIC CONCEPTS

Description:

The field of articulatory phonetics is concerned primarily with the question of how speech is realized through movements of body structures. While articulatory phoneticians have often described speech sounds using terms that refer to an inventory of body parts (e.g., “tongue,” “lips,” “velum,” etc.), a core challenge of articulatory phonetics is to understand how such structures function and interact to produce speech sounds. A complete understanding of articulatory phonetics thus requires that we define ways of mapping our descriptive terms onto groupings of nerves and muscles that our brains and bodies can use to produce speech movements. This chapter explores some of the ways articulatory phoneticians describe speech sounds, suggesting that a more fully embodied approach can provide novel insights into speech articulation and can help to understand links between speech and other functions of the human vocal tract.



Another way articulatory phoneticians have reduced the dimensionality of the vocal tract is by describing it in terms of a set of fixed, anatomically defined structures, often referred to using familiar body-part terms such as “lips” or “tongue tip.”

The lips are the most visible part of the vocal tract, making them an easy place to start thinking about how the body makes speech sounds. Lips are associated with the production of bilabial speech sounds such as /b/, /m/ and /w/ and with labio-dental sounds such as /f/ and /v/. Despite the distinct structures and functions of the upper lip and the lower lip, the English language reserves a single term (“lip”) for both structures, referring roughly to the fleshy parts immediately adjacent to the vermilion border (the often darker-hued bands immediately surrounding the mouth).

In contrast, Jahai, a Mon-Khmer language spoken in Malaysia, uses a separate word for each lip, with nus “upper lip” including all the fleshy parts between the nose and mouth and tnit “lower lip” including all the fleshy parts of the lower lip and chin (Burenhult, 5 Articulatory phonetics 2006). Interestingly, defined in this way, the Jahai “articulators” could be said to reflect more accurately the underlying innervation, with the nus innervated by the buccal branches of the facial nerve and the tnit serviced by the marginal mandibular branches of the facial nerve

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	-	-	1	1	1	1	1

CO1: The student will be able to explain the fundamentals and classification of speech sounds.

References:

1. Claudio Becchetti and Lucio Prina Ricotti, “Speech Recognition”, John Wiley and Sons, 1999.
2. Daniel Jurafsky and James H Martin, “Speech and Language Processing – An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, Pearson Education, 2002.